George Paul

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My MSc dissertation grew into a first-author paper with Prof. Michael Tipping on a new Bayesian shrinkage prior, now submitted to Communications in Statistics. My next piece of work is a theoretical note on the guarantees behind the method. In industry, I fine-tuned and evaluated vision-language models as part of a team whose work became a peer-reviewed EC3 2026 paper on engineering drawing layout detection. I am currently situated in IBM's AI and Analytics Team.

Publications & Research Experience

The Adaptive Elastic Horseshoe Prior **Feb. 2025 – Present**
First author, with Prof. Michael Tipping. Submitted to Communications in Statistics *University of Bath*

- **Problem**: Shrinkage priors fix their regularisation geometry in advance, so no existing method adapts when sparse and correlated structures appear in the same data. Building energy data, where I first met the problem, is exactly that: a few strong drivers alongside many weak, collinear ones.
- **Approach**: A single prior that learns the balance between sparse and grouped shrinkage from the data, by empirical Bayes, and then gives the exact closed-form posterior over the coefficients. Showed that the procedure fitting the prior is well-defined and converges locally at the point it reaches in practice, and checked this against observed runs.
- **Reliability**: Diagnosed a non-convergence in an earlier version of the fitting procedure, a period-2 oscillation that $\hat{R}$ could not detect because all chains started from the same unconverged point. Correcting it reversed the real-data result.
- **Findings**: Across simulation studies and a standard benchmark dataset, the prior adapts to correlation as intended and reduces error relative to the horseshoe when features form strongly correlated groups; assigning a separate sparse prior to irrelevant features cuts false positives roughly threefold. It does not help on purely sparse designs or when features approach the sample size, its uncertainty is conditional on plug-in hyperparameters, and the paper reports all three.
- **Future work**: The adaptive updates are not yet derived from a single objective, and posterior contraction is open. I am working on both in a follow-up theoretical note (in preparation).
- Began as the MSc dissertation (81%) and developed independently alongside full-time employment.

Benchmarking Deep Learning for Engineering Drawing Layout Detection **2025 – 2026**
Co-author, peer-reviewed at EC3 2026 (DOI: 10.35490/EC3.2026.305) *Buro Happold*

- **Problem**: Document layout models are trained and benchmarked almost entirely on text-centric documents, so their reported performance says little about engineering drawings, which have sparse line geometry, extreme aspect ratios and structure carried by spatial relationships. No prior work had quantified that gap.
- **Approach**: Worked with facade engineers to agree what should be labelled and to build an annotated dataset of real drawings, then ran a controlled comparison of five architectures. I designed the evaluation protocol and led the vision-language model arm, fine-tuning Qwen3-VL with parameter-efficient adaptation.
- **Findings**: The vision-language model achieved the highest F1 of the five (0.911), with strong semantic classification but weaker localisation and a latency that rules out real-time use. An ablation showed that models pre-trained on general documents performed worse than equivalent models without that pre-training.
- **Future work**: Testing how far these results transfer beyond a single engineering domain, through zero-shot tests of domain shift and few-shot measurement of how quickly models adapt.

Professional Experience

IBM **Oct. 2026 – Present**
Data Science Associate (Machine Learning) *London, UK*

- Applying machine learning within IBM's AI and Analytics department.

Buro Happold **Sep. 2025 – Sep. 2026**
Data Science Analyst *London, UK*

- **Risk modelling**: Built a geospatial risk-scoring system combining five years of time-series outage data with satellite-derived features, tuning the precision/recall operating point against the asymmetric cost of false positives versus missed failures.
- **Validation**: Developed model-validation pipelines in Python on Azure, with explicit checks for robustness to distribution shift.
- **Model fine-tuning and evaluation**: Fine-tuned and evaluated vision-language and language models (PyTorch, LoRA, SWIFT; Azure ML, Hugging Face APIs) to extract structured information from technical documents; owned the labelling framework and evaluation protocol, contributing to the EC3 2026 paper above.
- **AI Super User**: Guided adoption of AI tooling across the business, running internal sessions and advising colleagues from non-technical disciplines.

Green Place Assets **Jun. 2024 – Jul. 2025**
Data Science Intern *Remote*

– Designed a building-performance index combining time-series volatility, peer benchmarking and multivariate scoring; built gradient-boosting and random-forest models on tabular and geospatial data (out-of-sample $R^2 = 0.81$), validated by cross-validation.

Self-employed **Jul. 2018 – Jun. 2024**
Mathematics & Statistics Tutor *UK*

– Taught GCSE and A-Level mathematics and statistics over six years, explaining technical material to people meeting it for the first time.

Education

University of Bath **Sep. 2024 – Jan. 2026**
MSc Data Science, Distinction (overall 80.7%) *Bath, UK*

– Statistics 94%, Machine Learning 91%, Bayesian Machine Learning 88%, Research 81%. Statistical inference, probabilistic modelling, time-series analysis, deep learning and reinforcement learning.

– Dissertation supervised by Prof. Michael Tipping (see Publications & Research Experience).

University of Manchester **Sep. 2021 – Jun. 2024**
BSc Mathematics, 2:1 *Manchester, UK*

– Probability theory, graph theory, mathematical statistics, stochastic processes, numerical methods, linear algebra; scientific computing in Python and MATLAB.

Beechen Cliff School **2019 – 2021**
A-Levels *Bath, UK*

– Further Mathematics A*, Mathematics A*, Physics A*, Chemistry A.

Skills & Collaboration

- **Programming & tools**: Python (NumPy, SciPy, scikit-learn, statsmodels, PyTorch, JAX, Hugging Face), REST/model API integration, SQL, MATLAB, Git/GitHub, Azure ML, LaTeX.
- **Research collaboration**: Co-authored a peer-reviewed paper in a multi-disciplinary team, working with facade engineering experts to define the annotation scheme and cross-check labelling; delivered a reinforcement learning project (PPO agent) as part of a student team during the MSc.
- **Team sport**: Played football competitively throughout my life in China, across Europe and in Africa, in team environments where cooperation matters more than individual ability.